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Temporal Analysis of Climate Variables and NDVI of Wetlands Within the Kankakee River Watershed from 2000-2021

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Abstract

The Kankakee River watershed once supported approximately 400,000 acres of wetland habitat across northern Indiana and Illinois, but much of this habitat has been lost within the last 150 years. This previously expansive wetland is commonly referred to as the Grand Kankakee Marsh, or the Grand Marsh. In the late 19th century, much of the Grand Kankakee Marsh was drained and converted into agricultural land, while major stretches of the Kankakee River were channelized. The resulting scenario today is a watershed containing fragmented wetland habitat, a straightened river prone to flooding, and predominantly agricultural land use (Fraser & Bleuer, 1991).

The Normalized Difference Vegetation Index (NDVI) is a commonly used remote sensing technique which utilizes the near-infrared (NIR) and the red (R) wavelength bands obtained from multispectral satellite imagery. Multispectral satellite imagery has been captured repeatedly over the last few decades from various satellite observation missions, which allows for the analysis of NDVI over large spatial and temporal scales

(Huang et al., n.d.). Additionally, regional climate data in the Midwest United States has become available at daily, monthly, and annual scales (Purdue University & National Oceanic and Atmospheric Administration, 2023). Climate studies have shown that Winter and Spring precipitation, along with average temperatures have been increasing in Indiana and the Midwestern United States within the last century (Purdue University, 2018). This study analyzed the relationships between NDVI values of freshwater emergent and forested wetlands within the Kankakee River watershed with temperature and precipitation data between the years 2000 and 2021. Additionally, this paper provides an overview of the geological, ecological, and recent human history regarding the Kankakee River and surrounding watershed.

1. Introduction

The Normalized Difference Vegetation Index (NDVI) is calculated from the near-infrared (NIR) and the red (R) wavelength bands obtained from a multispectral image and using the following equation: NDVI = NIR - R/ NIR + R (**Figure 1 - Figure3).**

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}}$$

Figure 1 above shows the formula to calculate NDVI (Huang et al., 2020).

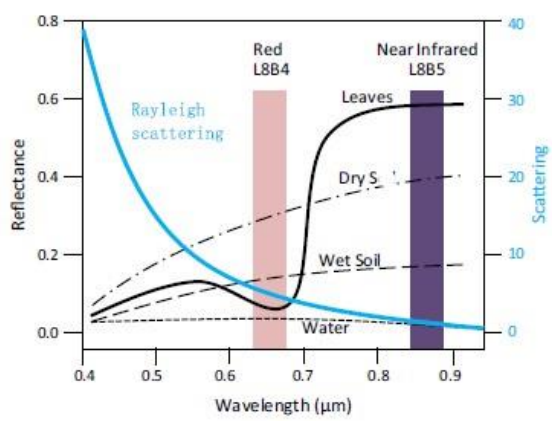


Fig. 1 An illustration of spectral response curves and their intersections with red (Landsat 8 Band 4) and near infrared bands (Landsat 8 Band 5) as well as the wavelength-dependent scattering

Figure 2 above shows the respective wavelengths of the red (Landsat 8 band 4) and near-infrared (Landsat 8 band 5) bands (Huang et al., 2020).

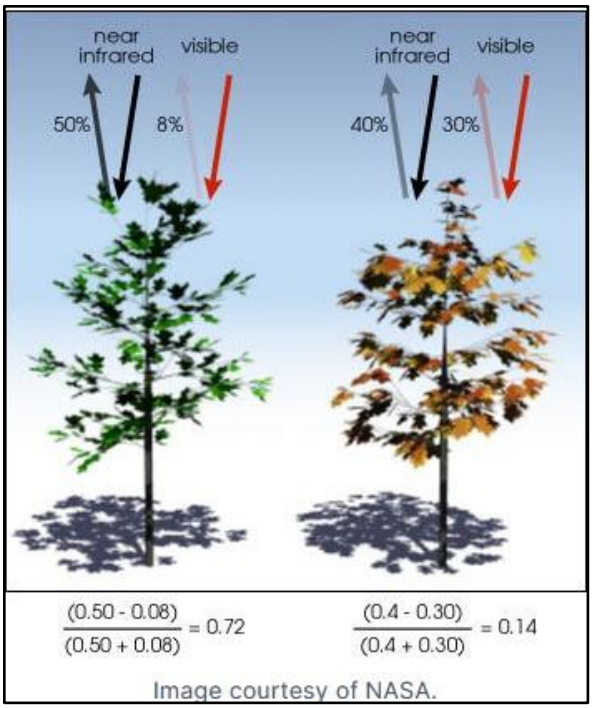
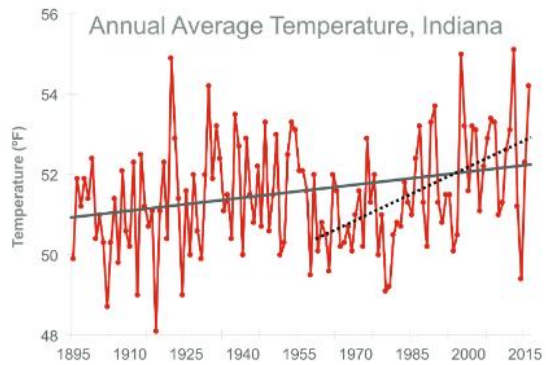


Figure 3 above shows an example of NDVI values based on spectral characteristics of vegetation at different stages of growth (Huang et al., 2020).

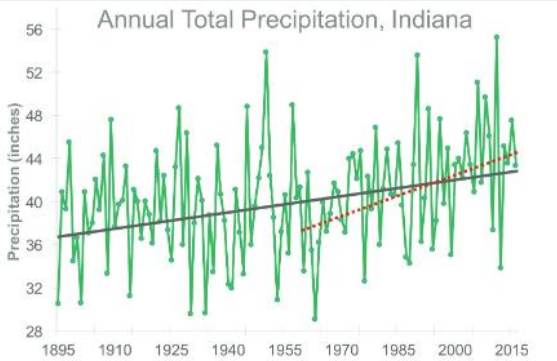
This index has proven to be useful and accurate for characterizing vegetation productivity (Zhang et al., 2020). Landsat 7 and Landsat 8 imagery have a moderate spatial resolution of 30 meters (m), meaning that each image pixel covers a 30 m by 30 m area, and a high temporal resolution, with full earth coverage being completed every 16 days (*USGS Landsat 8 Level 2, Collection 2, Tier 1 | Earth Engine Data Catalog*, n.d.).

Additionally, climate data for the Midwestern United States has been collected at daily, monthly, and annual scales by the Midwestern Regional Climate Center (MRCC) and includes temperature and precipitation data (Purdue University & National Oceanic and Atmospheric Administration, 2023). The relationship between these variables (NDVI, temperature, and precipitation) varies depending on regional climate and geographic location (Ichii, K; Kawabata, A; Yamaguchi, Y).

Since 1895, annual average temperature and annual total precipitation in Indiana have steadily increased, with a more positive trend of increase being observed since the late 1950s (**Figure 4 and Figure 5)** (Purdue University, 2018).



Above: *Statewide annual average temperature for Indiana from 1895 to 2016 is shown in red. The black solid line shows the increasing trend in annual temperature (0.1°F/decade) for the period from 1895 to 2016. The black dotted line shows the temperature trend since 1960 (0.4°F"/decade). Source: NOAA Climate at a Glance Database.*



Above: *Statewide annual total precipitation for Indiana from 1895 to 2016. Black solid line shows the increasing trend in annual precipitation (0.46"/decade) for the period from 1895 to 2016. The red dotted line shows the precipitation trend since 1960 (1.33"/decade). Source: NOAA Climate at a Glance Database.*

Figures 4 (top) and **5** (bottom) obtained from Purdue Climate Report (Purdue University, 2018).

In recent decades, changes to the distribution of annual precipitation in Indiana and the Midwest US have also been observed. Recorded weather data has shown that precipitation levels are increasing in the Spring months, and that Winter precipitation is more frequently coming in the form of rain rather than snow (Purdue University, 2018).

1.1 Geological and Ecological History of the Kankakee River Watershed

The historical expanse of the Grand Kankakee Marsh, which was a contiguous marsh and swamp habitat that spanned over 400,000 acres within the Kankakee River watershed, once represented a significant ecological feature of the midwestern landscape (**Figure 6)**.

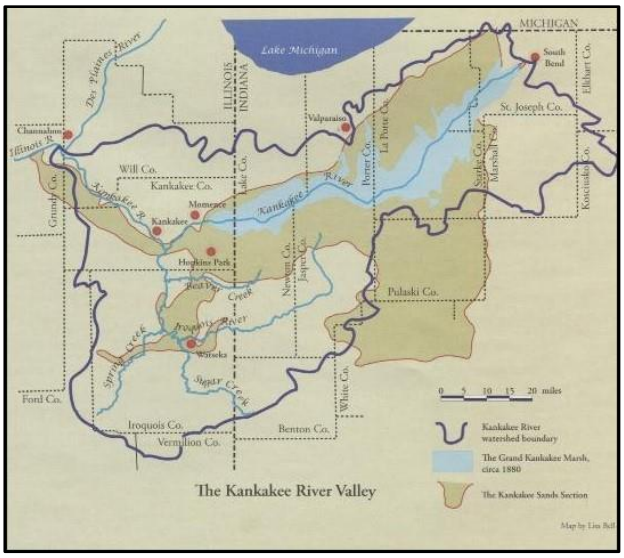


Figure 6 above shows the historical extent of the Grand Kankakee Marsh.

During the late Pleistocene Epoch, glaciers advanced into Indiana several different times. The most recent stage of glaciation in this area was the Wisconsin Stage of Glaciation, which lasted from 24,000 to 10,000 years ago (State of Indiana, Department of Natural Resources, Division of Water, 1990). During this time, much of the Kankakee basin was covered in ice. Around 19,000 years ago, a series of flood events resulting from the breaching of moraines surrounding Glacial Lake Chicago (**Figure 7)**, flooded the Kankakee River basin and shaped the modern geomorphology of the river. These flood events are referred to as the Kankakee Torrent (United States Geological Survey & Leverett, 1899)

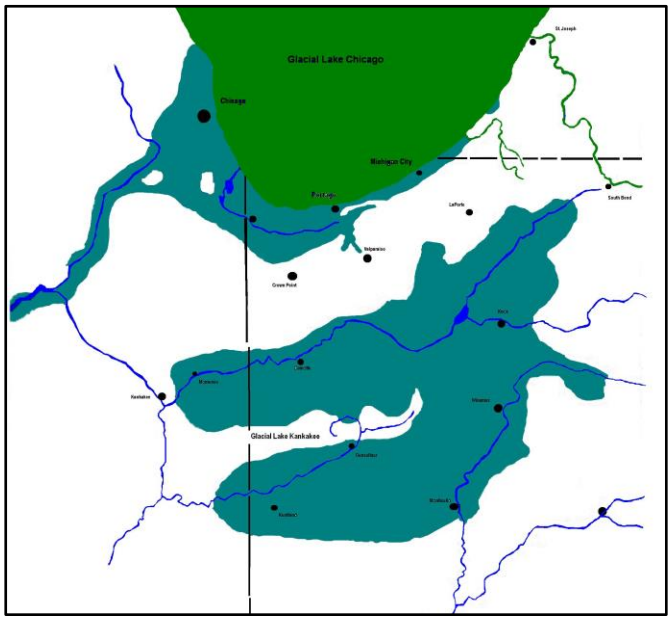


Figure 7 above shows the lower expanse of Glacial Lake Chicago, which is thought to be the source for the flood events during the Kankakee Torrent (United States Geological Survey & Leverett, 1899).

Around 15,000 years ago, the Michigan lobe to the north and the Saginaw lobe to the east retreated, leaving the Kankakee basin and Kankakee River free of ice, while other surrounding areas remained covered in ice (Curry et al., 2014) (**Figure 8 and Figure 9)**.

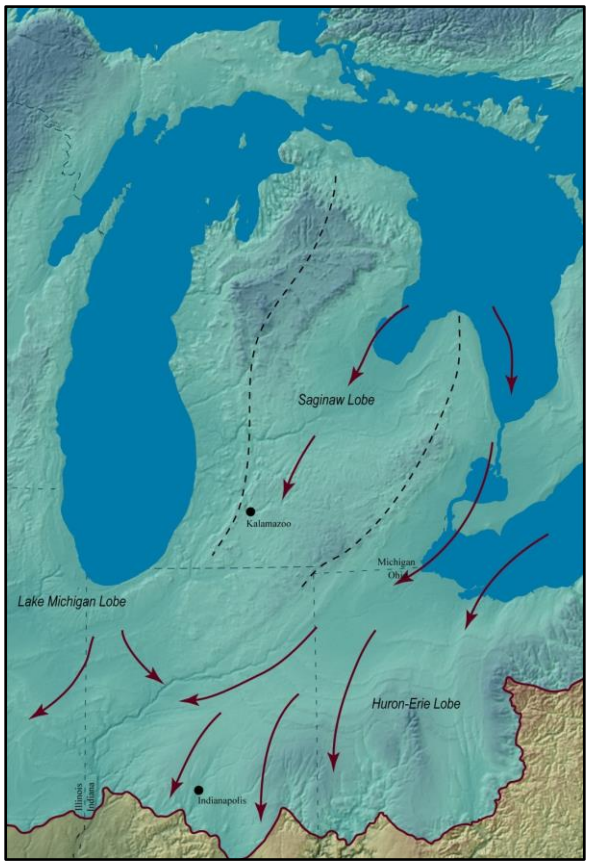


Figure 8 above illustrates the extent of the Wisconsin glaciation across Indiana, which completely covered the Kankakee Watershed. Source: Indiana Geological Survey

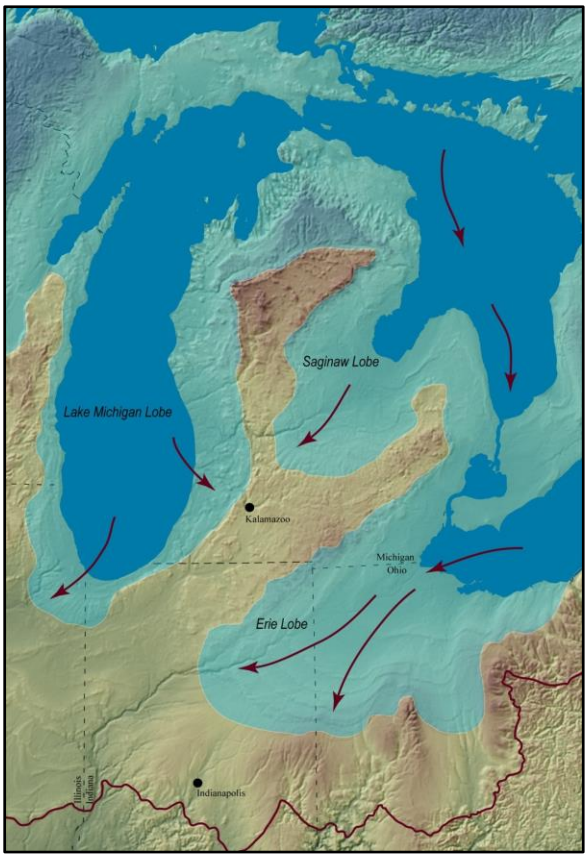


Figure 9 above illustrates the extent of glaciation toward the end of the Wisconsin episode, which no longer covered the Kankakee Watershed in ice. The southern Michigan Lobe sits at the boundary of the Valparaiso Moraine. Source: Indiana Geological Survey

To the north of the Kankakee basin lay a landform known as the Valparaiso Moraine, which consisted of glacial till and sand. This moraine served as a glacial boundary for the Michigan Lobe during the Wisconsin period. To the south of the Kankakee Basin lay another moraine known as the Iroquois Moraine (**Figure 10)**.

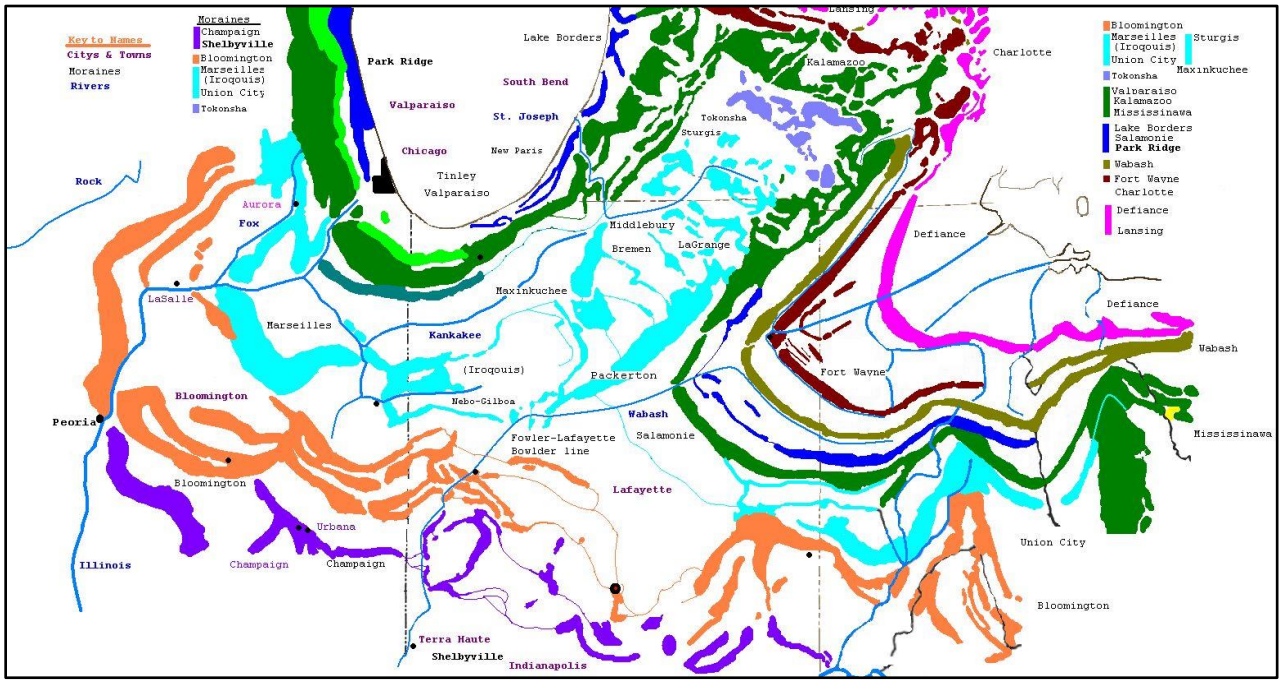


Figure 10 above shows the locations of the Valparaiso Moraine (green color around Lake Michigan) and the Iroquois Moraine (blue color south of Kankakee River) (Light et al., 2014).

To the west of the basin, is a large limestone ridge. Together, the Valparaiso Moraine, Iroquois Moraine, and the limestone ridge to the west provided the topographic features to trap water in the Kankakee basin. When this region experienced seasonal warming, glacial meltwater was discharged from the Michigan Lobe, which carried clay and sand into the Kankakee basin. Eventually, the basin filled with water and became a large lake (State of Indiana, Department of Natural Resources, Division of Water, 1990), which will be referred to as Lake Kankakee in this paper.

Beginning approximately 11,000 years ago, the Wisconsin glaciation began to permanently retreat. Due to this, Lake Kankakee no longer received seasonal meltwater from surrounding glaciers. This allowed Lake Kankakee to drain, while filling in with deposited sand and other sediments. This began the formation of the Grand Kankakee Marsh, surrounded by a landscape that today is referred to as the Kankakee Outwash Plain. The Kankakee Outwash Plain consists of a smooth, flat, sandy plain with remnants of sand dunes (Fraser & Bleuer, 1991) (**Figure 11)**.

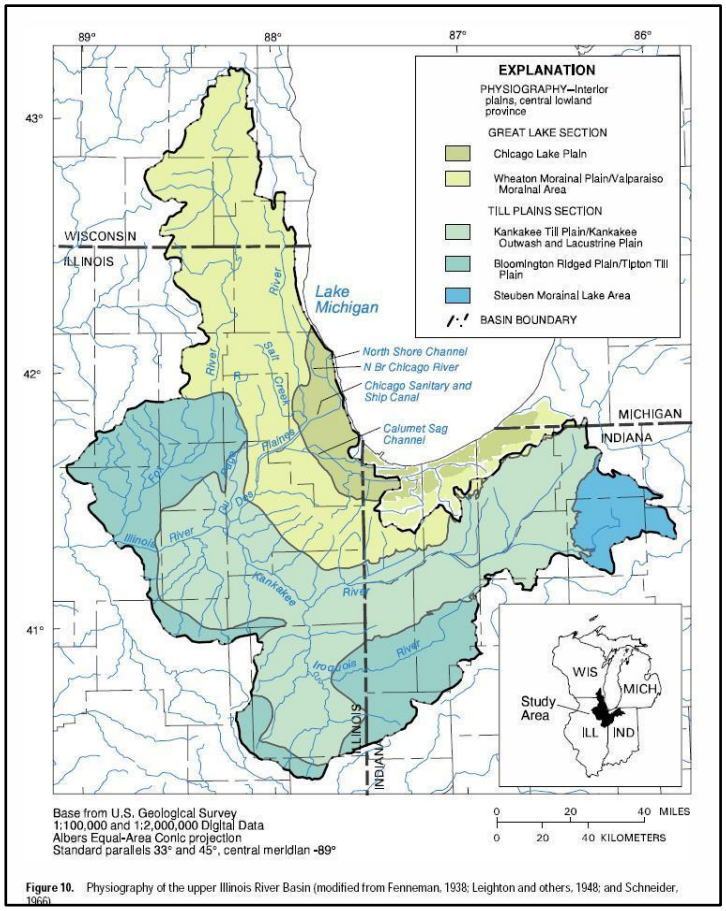


Figure 11 above shows the Kankakee Outwash Plain. Source: US Geological Survey.

1.2 Legal History of Kankakee River Channelization and Wetland Development

In 1850, the Swamp Lands Act was passed by the US Congress, which allowed for the sale of federally owned swamp land to private owners, provided that the land was then drained for agricultural or commercial purposes. By 1854, the federal

government approved 9,811,682 acres as "wet" lands within Indiana which were eligible for sale and conversion (Strausberg, 1977). By the late 1880s, Indiana began enacting a large-scale plan to dredge, straighten, and channelize the Kankakee River, which was largely completed by the early 1920s. This left a straight channel on the Indiana side to be met with an abrupt return to a more natural, sinuous river on the Illinois side (Ivens et al., 1981) (**Figure 12, Figure 13, and Figure 14)**.

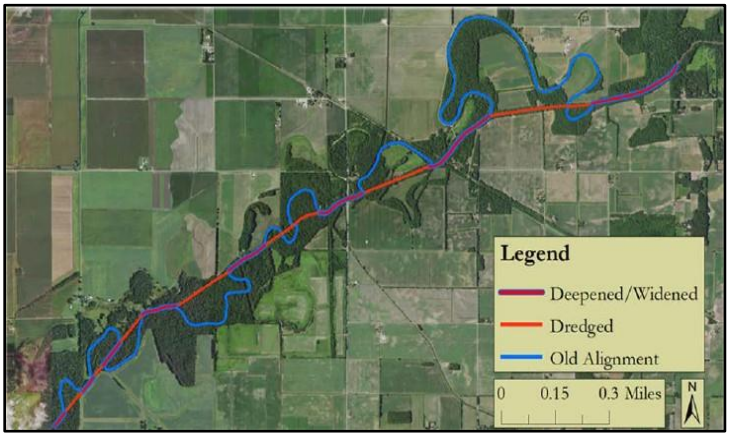


Figure 12 above shows the original Kankakee River alignment (blue), stretches of the river that were dredged (red), and stretches that were deepened and widened (red with blue outline) (Christopher B. Burke Engineering, LLC, 2019).

Figure 13 above shows steam dredge equipment that was used to dredge and channelize the Kankakee River (*Dredging Of The River*, n.d.).

Figure 14 above shows a channelized portion of the Kankakee River surrounded by corn fields near Crumstown, Indiana. Photo Credit: Chris Light (Light, 2008).

The combination of draining these wetlands and channelizing the river resulted in the hydrological and ecological scenario for the watershed that exists today.

2. Method

2.1 Study Area

The Kankakee River flows westward from Indiana into Illinois and has a drainage basin area of approximately 5,165 square

miles, with 2,996 square miles being within Indiana. The river has a total length of approximately 150 miles, with 91 miles being within Indiana (Ivens et al., 1981). This study focused on the HUC 8 watershed (HUC 07120001) of the Kankakee River, which encompasses an area measuring 1,938,667 acres (**Figure 15)**.

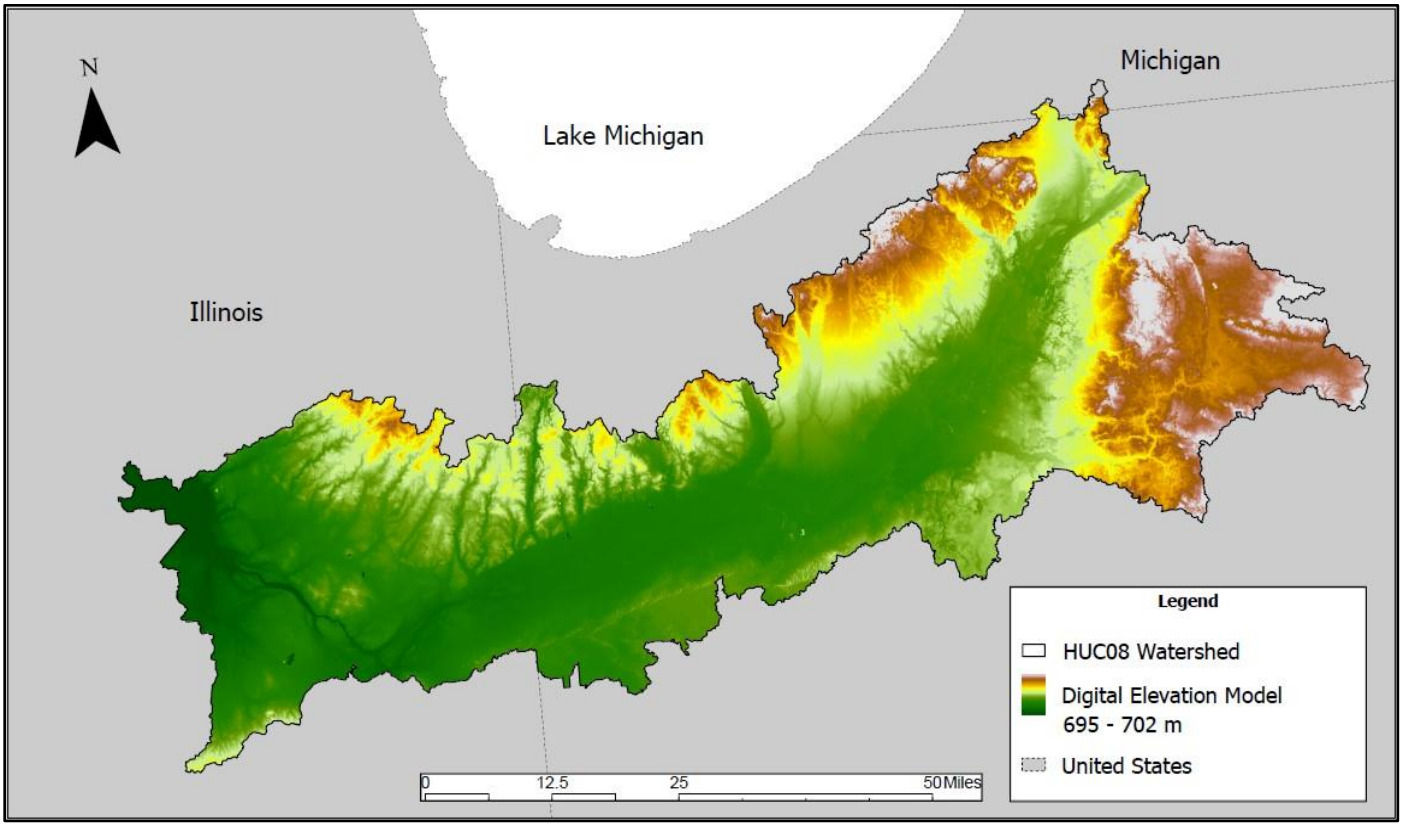


Figure 15 above shows a Digital Elevation Model (DEM) of the HUC 8 watershed of the Kankakee River. Elevation values range from 695 m – 702 m. High elevation areas are displayed in orange, yellow, and white. Lower elevations are displayed in shades of green. Source: Prairie Research Institute (Illinois) and Indiana Department of Natural Resources. Image mosaic and map created by Ben Grubbs (Prairie Research Institute ArcGIS Hub, 2018) (Indiana Department of Natural Resources ArcGIS Online, 2020).

The HUC 8 watershed contains 127,261 acres of mapped wetland habitat across

25,150 polygons. These polygons consist of freshwater emergent, freshwater forested, riverine, lake, and pond wetland types. This study focused on freshwater emergent and forested wetlands. Freshwater emergent wetlands consisted of 33,982 acres of wetland habitat across 8,249 polygons, while forested wetlands consisted of 52,397.1 acres across 7,767 polygons (U.S. Fish and Wildlife Service, 2023) (**Figure 16).**

Figure 16 below shows a pie chart representing the relative proportion and acreage of wetland types within the HUC 8 watershed.

2.2 Data collection

The general workflow for this study is outlined in **Figure 17**.

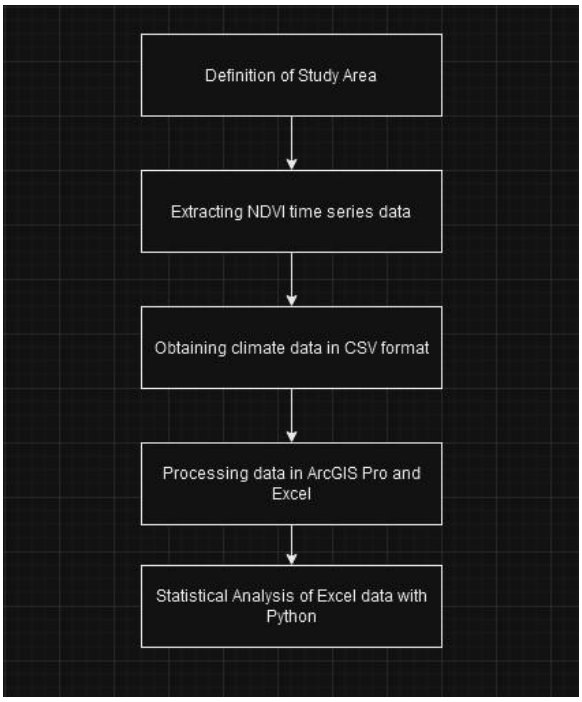
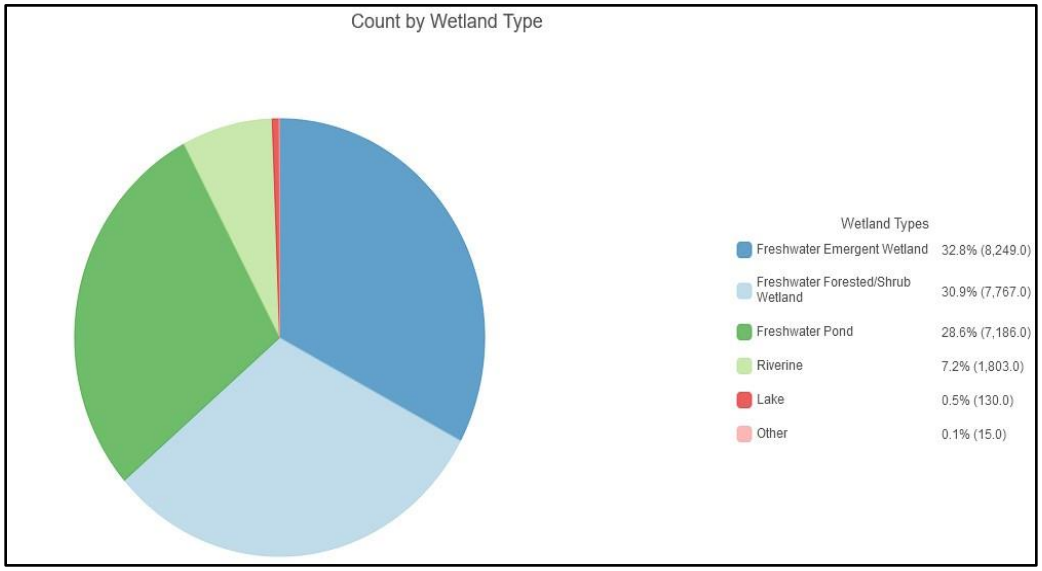


Figure 17 above shows a flowchart outlining the general steps involved for this study.



A shapefile containing all mapped wetlands within the HUC 08 watershed of the Kankakee River was obtained from the National Wetlands Inventory database (U.S. Fish and Wildlife Service, 2023). This shapefile was uploaded to Google Earth Engine (GEE) to be used as the project area for analysis. Using the JavaScript code editor, multiple scripts were developed to import Landsat 7 and Landsat 8 Collection 2 Tier 1 surface reflectance data, clip the data within the wetland polygons, calculate mean NDVI, and export a CSV file containing the results for individual years ranging from 2000 to 2021. When doing this, mean NDVI values were calculated within the month of August to represent peak vegetation growth (**Figure 18)**.

NDVI Spreadsheet Landsat 8)

```
Imports (2 entries)
  var roi: Polygon, 6 vertices
  var wetlands: Table projects/ee-bcgrubbs/assets/Wetlands
// Load Landsat 8 surface reflectance data
var collection = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2')
  .filterDate('2014-08-01', '2014-08-31')
  .filterBounds(roi);

// Define a function to mask clouds and shadows using QA_PIXEL band (Collection 2)
function maskClouds(image) {
  // Bits 4 and 3 are cloud shadow and cloud, respectively.
  var cloudShadowBitMask = (1 << 4); // Updated for Collection 2
  var cloudsBitMask = (1 << 3); // Updated for Collection 2
  // Get the pixel QA band.
  var qa = image.select('QA_PIXEL');
  // Both flags in the QA band should be set to zero, indicating clear conditions.
  var mask = qa.bitwiseAnd(cloudShadowBitMask).eq(0)
    .and(qa.bitwiseAnd(cloudsBitMask).eq(0));
  return image.updateMask(mask);
}

// Apply the cloud masking function to the image collection
var collectionMasked = collection.map(maskClouds);

// Define a function to calculate NDVI
function addNDVI(image) {
  var ndvi = image.normalizedDifference(['SR_B5', 'SR_B4']).rename('NDVI');
  return image.addBands(ndvi);
}

// Apply the function to the masked image collection
var collectionNDVI = collectionMasked.map(addNDVI);

// Load the shapefile and calculate the mean NDVI values within its polygons
var shapefile = ee.FeatureCollection('projects/ee-bcgrubbs/assets/Wetlands');
var meanNDVI = collectionNDVI.select('NDVI').mean()
  .reduceRegions({
    collection: shapefile,
    reducer: ee.Reducer.mean(),
    scale: 30,
  });

// // Export the mean NDVI values to a Google Drive spreadsheet
// Export.table.toDrive({
//   collection: meanNDVI,
//   description: 'L8',
//   folder: 'Landsat 8',
//   fileFormat: 'CSV',
// });

Map.addLayer(meanNDVI, {min: -1, max: 1, palette: ['blue', 'white', 'green']}, 'Mean NDVI');
```

Figure 18 above contains JavaScript code used to calculate and export mean NDVI values within the study area using Google Earth Engine (GEE). Source: Ben Grubbs and GEE.

The original shapefile, along with each spreadsheet of NDVI data previously exported from GEE were imported to ArcGIS Pro and merged into a single data set using the “Spatial Join” tool. The joined shapefile was then filtered to only include results for freshwater emergent or freshwater forested wetlands, and blank values were removed. Once filtered, the NDVI results were

exported into CSV files for statistical analysis using Python.

Climate data were obtained from the Midwestern Regional Climate Center (MRCC) (Purdue University & National Oceanic and Atmospheric Administration, 2023). The study area spanned across 4 climate regions, respectively referred to as Climate Region 1 in Indiana (CR 1), Climate Region 2 in Indiana (CR 2), Climate Region 3 in Illinois (CR 3), and Climate Region 4 in Illinois (CR 4) (**Figure 19)**.

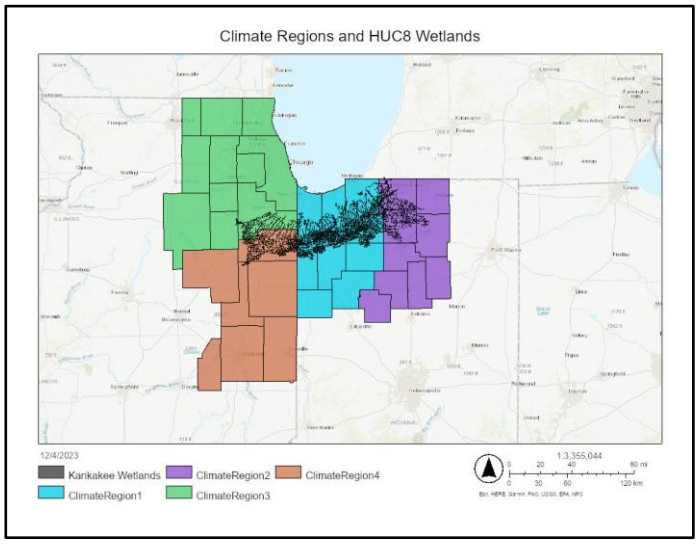


Figure 19 above shows Climate Region 1 (blue), Climate Region 2 (purple), Climate Region 3 (green), and Climate Region 4 (orange). Map created by Ben Grubbs using ArcGIS Online.

Climate data for monthly precipitation and temperature were downloaded in CSV format for each climate region spanning from 2000 to 2021. To improve representation of NDVI results, the wetlands shapefile was clipped into four new shapefiles to represent the corresponding climate regions. The new shapefile data was exported as CSV to later be cleaned and combined with the climate region datasets.

2.3 Data Analysis

The programming language Python was used for data analysis. The Python libraries known as “Pandas”, “SciPy”, “Matplotlib”, and “statsmodels” were used for spreadsheet editing, statistical analysis, and data visualization. All Python scripts were run using the Visual Studio Code application.

A Python script was used to reorganize the data and to calculate summary statistics of the NDVI data, which

gave a mean NDVI value for all freshwater emergent and freshwater forested wetland polygons for each year from 2000 through 2021 within each of the four climate regions.

Year	R4emergentNDVI	R4AnnualPrecip	R4AvgSummerTemp	R4WinterSpringPrecip	R4AvgAnnual	R4AvgWinterSpringTemp	R4ForestedNDVI
2000	0.370	34.43	69.68	11.69	50.87	43.7	0.363401453
2001	0.317	37.38	70.23	11.39	52.32	42.18	0.276531489
2002	0.164	37.04	70.28	18.18	51.97	42.28	0.164889553
2003	0.339	38.64	68.48	11.14	50.44	39.64	0.359618623
2004	-0.011	40.02	68.43	13.95	51.5	42.72	-0.016631662
2005	0.296	33.34	71.03	13.21	52.42	42.6	0.248463579
2006	0.291	41.31	69.98	15.32	52.95	44.82	0.266533118
2007	0.274	39.66	71.83	14.47	52.3	41.8	0.263560255
2008	0.367	48.3	68.58	18.27	49.85	39.24	0.354940258
2009	0.380	46.65	68.18	18.33	50.23	40.58	0.373707753
2010	0.245	36.24	72.48	12.51	51.64	41.62	0.259087254
2011	0.339	39.73	71.35	18.32	51.93	40.48	0.305618971
2012	0.292	30.54	72.75	10.18	54.59	47.9	0.289576695
2013	0.342	36.99	69.55	19.17	50.18	40.62	0.340530538
2014	0.419	41.52	69.1	13.4	48.21	36.22	0.413451505
2015	0.347	47.52	69.7	12.23	51.58	38.9	0.384352266
2016	0.448	40.46	70.95	11.86	53.68	43.38	0.411654536
2017	0.403	38.41	69.25	17.51	53.34	45.78	0.388462854
2018	0.413	40.7	72.9	13.74	51.33	40.62	0.398416888
2019	0.357	43.1	70	20.33	50.91	39.98	0.390611667
2020	0.412	38.68	70.08	19.07	52.22	42.34	0.387295479
2021	0.425	42.99	70.43	13.63	52.7	40.84	0.402493723

For each year, NDVI values were averaged across all polygons within that region to obtain the final NDVI value for that year. The data was then reorganized into four new spreadsheets (one for each climate region), each containing a column representing the year, emergent wetland NDVI, forested wetland NDVI, annual precipitation, average summer temperature, Winter and Spring cumulative precipitation, average annual temperature, and Average Winter and Spring temperature (**Figure 20)**.

Figure 20 above shows the spreadsheet of final aggregated variables for region 4 that were used for statistical analysis.

Once the four new spreadsheets were organized, they were each imported into a Python script which was run to obtain a Pearson's Correlation Coefficient and p-value (**Figure 21)**.

```python
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm

from scipy.stats import pearsonr

# Read the CSV file
df = pd.read_excel(r"C:\Users\benja\OneDrive\Desktop\OSU\GEOG-560\Project Data\Data Analysis\R4_merged.xlsx")

#print df
print(df)
```

```python
#Freshwater Forested

# Calculate Pearson correlation between NDVI and Average Summer Temperature
correlation_ndvi_summer, p_value_ndvi_summer = pearsonr(df['R4ForestedNDVI'], df['R4AvgSummerTemp'])
print(f"Pearson correlation between R4 Freshwater Forested NDVI and Average Summer Temperature: {correlation_ndvi_summer:.4f}, p-value: {p_value_ndvi_summer:.4f}")

# Calculate Pearson correlation between NDVI and Average Winter-Spring Temperature
correlation_ndvi_winter_spring, p_value_ndvi_winter_spring = pearsonr(df['R4ForestedNDVI'], df['R4AvgWinterSpringTemp'])
print(f"Pearson correlation between R4 Freshwater Forested NDVI and Average Winter-Spring Temperature: {correlation_ndvi_winter_spring:.4f}, p-value: {p_value_ndvi_winter_spring:.4f}")

# Calculate Pearson correlation between NDVI and Average Annual Temperature
correlation_ndvi_annual, p_value_ndvi_annual = pearsonr(df['R4ForestedNDVI'], df['R4AvgTemp'])
print(f"Pearson correlation between R4 Freshwater Forested NDVI and Average Annual Temperature: {correlation_ndvi_annual:.4f}, p-value: {p_value_ndvi_annual:.4f}")

# Calculate Pearson correlation between NDVI and Cumulative Winter-Spring Precipitation
correlation_ndvi_winter_spring_precip, p_value_ndvi_winter_spring_precip = pearsonr(df['R4ForestedNDVI'], df['R4WinterSpringPrecip'])
print(f"Pearson correlation between R4 Freshwater Forested NDVI and Cumulative Winter-Spring Precipitation: {correlation_ndvi_winter_spring_precip:.4f}, p-value: {p_value_ndvi_winter_spring_precip:.4f}")

# Calculate Pearson correlation between NDVI and Annual Precipitation
correlation_ndvi_annual_precip, p_value_ndvi_annual_precip = pearsonr(df['R4ForestedNDVI'], df['R4AnnualPrecip'])
print(f"Pearson correlation between R4 Freshwater Forested NDVI and Precipitation: {correlation_ndvi_annual_precip:.4f}, p-value: {p_value_ndvi_annual_precip:.4f}")
```

```python
#Freshwater Emergent

# Calculate Pearson correlation between NDVI and Average Summer Temperature
correlation_ndvi_summer, p_value_ndvi_summer = pearsonr(df['R4meanNDVI'], df['R4AvgSummerTemp'])
print(f"Pearson correlation between R4 Freshwater Emergent NDVI and Average Summer Temperature: {correlation_ndvi_summer:.4f}, p-value: {p_value_ndvi_summer:.4f}")

# Calculate Pearson correlation between NDVI and Average Winter-Spring Temperature
correlation_ndvi_winter_spring, p_value_ndvi_winter_spring = pearsonr(df['R4meanNDVI'], df['R4AvgWinterSpringTemp'])
print(f"Pearson correlation between R4 Freshwater Emergent NDVI and Average Winter-Spring Temperature: {correlation_ndvi_winter_spring:.4f}, p-value: {p_value_ndvi_winter_spring:.4f}")

# Calculate Pearson correlation between NDVI and Average Annual Temperature
correlation_ndvi_annual, p_value_ndvi_annual = pearsonr(df['R4meanNDVI'], df['R4AvgTemp'])
print(f"Pearson correlation between R4 Freshwater Emergent NDVI and Average Annual Temperature: {correlation_ndvi_annual:.4f}, p-value: {p_value_ndvi_annual:.4f}")

# Calculate Pearson correlation between NDVI and Cumulative Winter-Spring Precipitation
correlation_ndvi_winter_spring_precip, p_value_ndvi_winter_spring_precip = pearsonr(df['R4meanNDVI'], df['R4WinterSpringPrecip'])
print(f"Pearson correlation between R4 Freshwater Emergent NDVI and Cumulative Winter-Spring Precipitation: {correlation_ndvi_winter_spring_precip:.4f}, p-value: {p_value_ndvi_winter_spring_precip:.4f}")

# Calculate Pearson correlation between NDVI and Annual Precipitation
correlation_ndvi_annual_precip, p_value_ndvi_annual_precip = pearsonr(df['R4meanNDVI'], df['R4AnnualPrecip'])
print(f"Pearson correlation between R4 Freshwater Emergent NDVI and Precipitation: {correlation_ndvi_annual_precip:.4f}, p-value: {p_value_ndvi_annual_precip:.4f}")
```

```python
#Temperature and Precipitation

# Calculate Pearson correlation between Winter-Spring Precipitation and Average Winter-Spring Temperature
correlation_winter_spring_precip_temp, p_value_winter_spring_precip_temp = pearsonr(df['R4WinterSpringPrecip'], df['R4AvgWinterSpringTemp'])
print(f"Pearson correlation between R4 Winter-Spring Precipitation and Average Winter-Spring Temperature: {correlation_winter_spring_precip_temp:.4f}, p-value: {p_value_winter_spring_precip_temp:.4f}")

# Calculate Pearson correlation between Annual Precipitation and Average Annual Temperature
correlation_annual_precip_temp, p_value_annual_precip_temp = pearsonr(df['R4AnnualPrecip'], df['R4AvgTemp'])
print(f"Pearson correlation between R4 Annual Precipitation and Average Annual Temperature: {correlation_annual_precip_temp:.4f}, p-value: {p_value_annual_precip_temp:.4f}")
```

Figure 21 above contains the Python script used for statistical analysis.

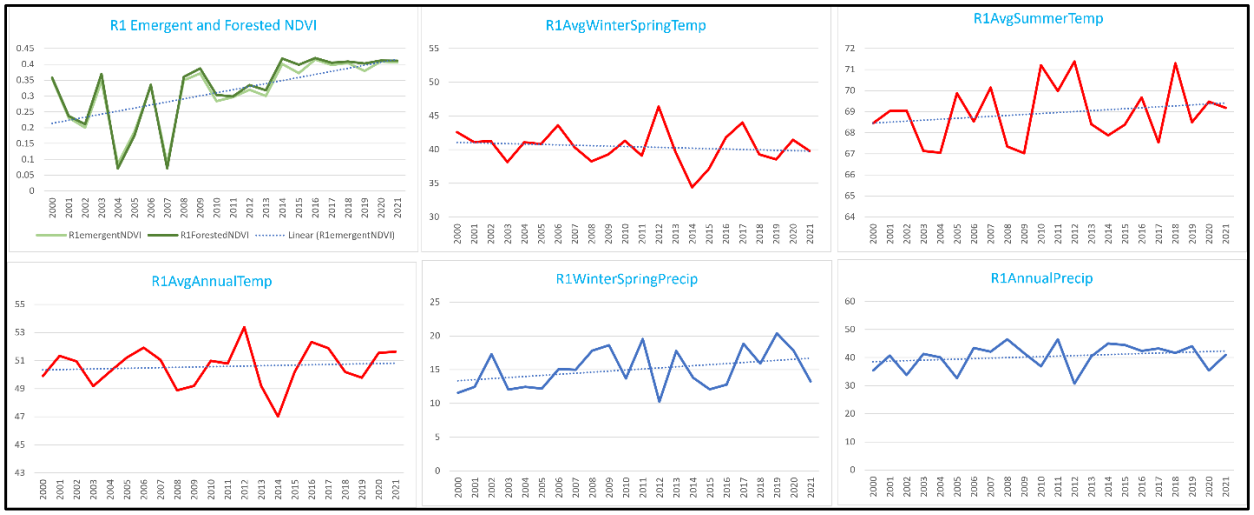
Results:

This study analyzed the relationships between mean NDVI values observed in August of each year for freshwater emergent wetlands and forested wetlands with the following variables: average summer temperature, average Winter and Spring temperature, average annual temperature, cumulative Winter and Spring precipitation, and annual precipitation.

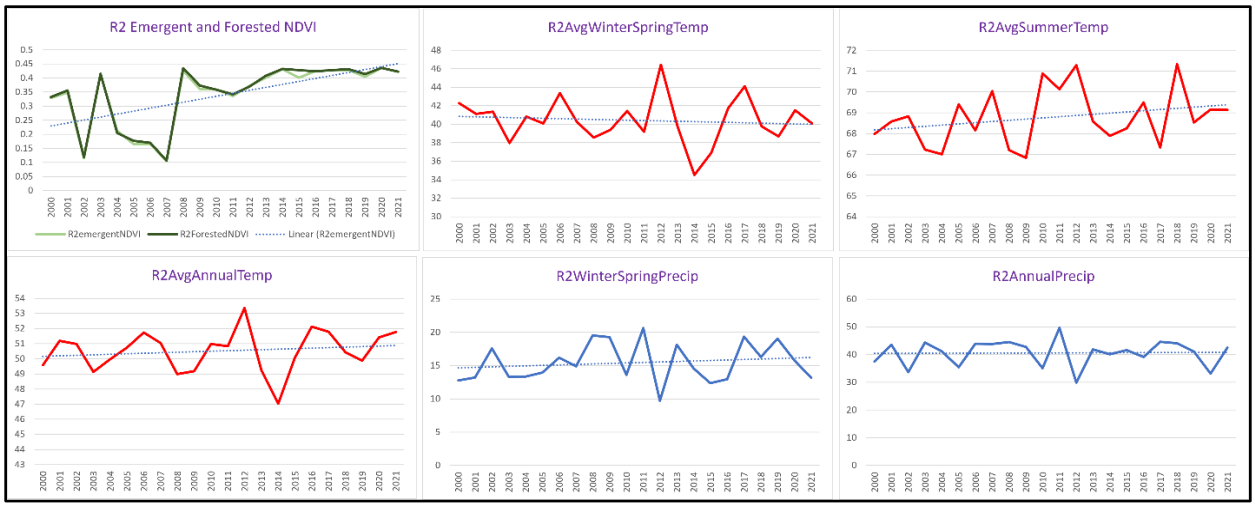
Additionally, the relationship between average Winter and Spring Precipitation and average Winter and Spring temperature were analyzed, along with annual precipitation and average annual temperature.

Figures 22 through **30** contain tables showing the statistical results for each region. The tables show the climate region, wetland type, variable 1, variable 2, Pearson Correlation (r) value, p-value, correlation type (positive/negative), and whether the results were statistically significant (if p value was less than 0.05).

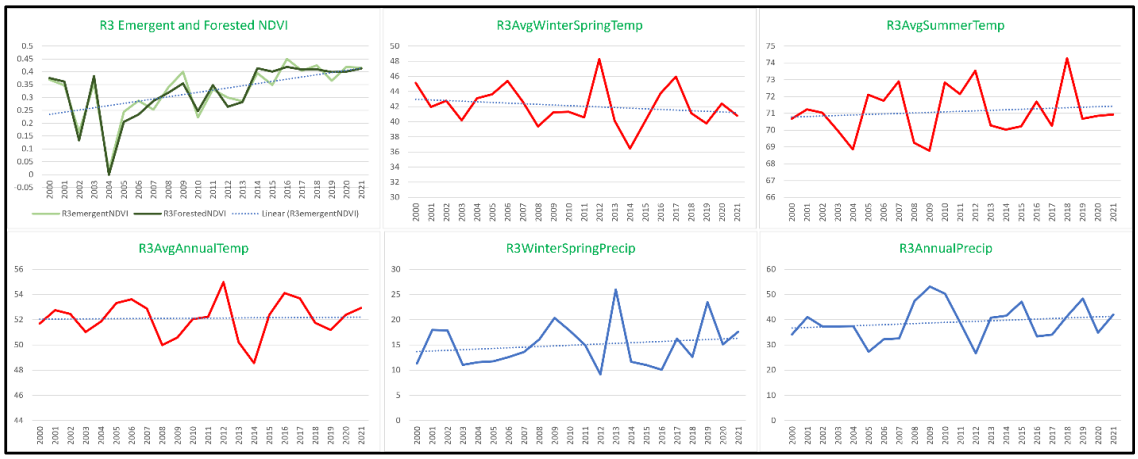
Region	Wetland Type	variable 1	variable 2	Pearson Correlation	p-value	Correlation	Significance
1	Freshwater Forested	mean NDVI	Average Summer Temperature	-0.0877	0.698	Weak Negative	Not Significant
1	Freshwater Forested	mean NDVI	Average Winter-Spring Temperature	-0.1805	0.4216	Weak Negative	Not Significant
1	Freshwater Forested	mean NDVI	Average Annual Temperature	-0.1417	0.5294	Weak Negative	Not Significant
1	Freshwater Forested	mean NDVI	Cumulative Winter-Spring Precipitation	0.1966	0.3805	Weak Positive	Not Significant
1	Freshwater Forested	mean NDVI	Annual Precipitation	0.2719	0.229	Weak Positive	Not Significant
1	Freshwater Emergent	mean NDVI	Average Summer Temperature	-0.0752	0.7393	Weak Negative	Not Significant
1	Freshwater Emergent	mean NDVI	Average Winter-Spring Temperature	-0.1565	0.4867	Weak Negative	Not Significant
1	Freshwater Emergent	mean NDVI	Average Annual Temperature	-0.1136	0.6148	Weak Negative	Not Significant
1	Freshwater Emergent	mean NDVI	Cumulative Winter-Spring Precipitation	0.1952	0.3841	Weak Positive	Not Significant
1	Freshwater Emergent	mean NDVI	Annual Precipitation	0.2709	0.2227	Weak Positive	Not Significant
1	N/A	Winter-Spring Precipitation	Average Winter-Spring Temperature	-0.19	0.397	Weak Negative	Not Significant
1	N/A	Annual Precipitation	Average Annual Temperature	-0.4429	0.039	Moderate Negative	Significant



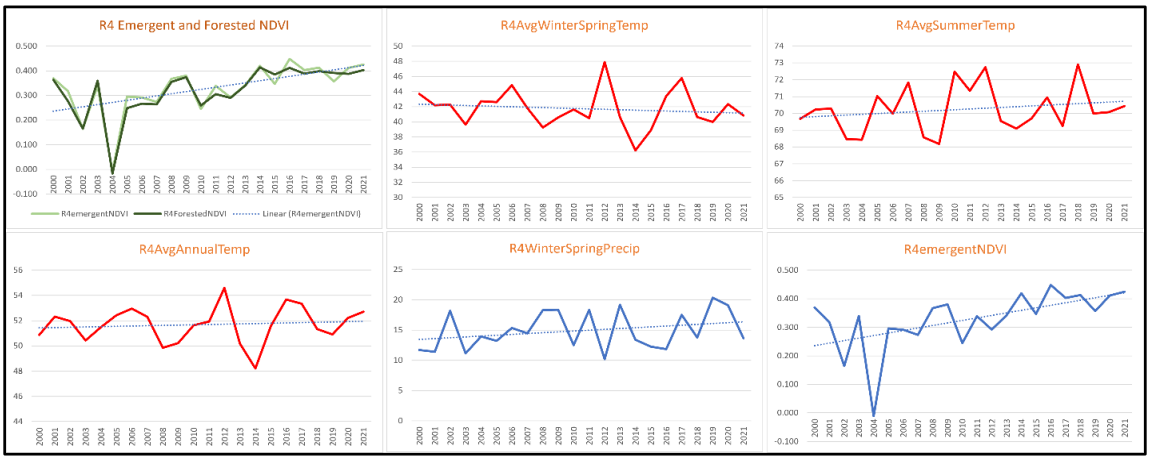
Region	Wetland Type	variable 1	variable 2	Pearson Correlation	p-value	Correlation	Significance
2	Freshwater Forested	mean NDVI	Average Summer Temperature	-0.0647	0.7747	Weak Negative	Not Significant
2	Freshwater Forested	mean NDVI	Average Winter-Spring Temperature	-0.2506	0.2606	Weak Negative	Not Significant
2	Freshwater Forested	mean NDVI	Average Annual Temperature	-0.2031	0.3646	Weak Negative	Not Significant
2	Freshwater Forested	mean NDVI	Cumulative Winter-Spring Precipitation	0.0533	0.8137	Weak Positive	Not Significant
2	Freshwater Forested	mean NDVI	Annual Precipitation	0.1385	0.5388	Weak Positive	Not Significant
2	Freshwater Emergent	mean NDVI	Average Summer Temperature	-0.052	0.8182	Weak Negative	Not Significant
2	Freshwater Emergent	mean NDVI	Average Winter-Spring Temperature	-0.224	0.3162	Weak Negative	Not Significant
2	Freshwater Emergent	mean NDVI	Average Annual Temperature	-0.187	0.4048	Weak Negative	Not Significant
2	Freshwater Emergent	mean NDVI	Cumulative Winter-Spring Precipitation	0.0398	0.8604	Weak Positive	Not Significant
2	Freshwater Emergent	mean NDVI	Annual Precipitation	0.1238	0.5831	Weak Positive	Not Significant
2	N/A	Winter-Spring Precipitation	Average Winter-Spring Temperature	-0.2082	0.3525	Weak Negative	Not Significant
2	N/A	Annual Precipitation	Average Annual Temperature	-0.3024	0.1713	Weak Negative	Not Significant



Region	Wetland Type	variable 1	variable 2	Pearson Correlation	p-value	Correlation	Significance
3	Freshwater Forested	mean NDVI	Average Summer Temperature	0.049	0.8286	Weak Positive	Not Significant
3	Freshwater Forested	mean NDVI	Average Winter-Spring Temperature	-0.2926	0.1864	Weak Negative	Not Significant
3	Freshwater Forested	mean NDVI	Average Annual Temperature	-0.1362	0.5455	Weak Negative	Not Significant
3	Freshwater Forested	mean NDVI	Cumulative Winter-Spring Precipitation	0.0452	0.8416	Weak Positive	Not Significant
3	Freshwater Forested	mean NDVI	Annual Precipitation	0.234	0.2947	Weak Positive	Not Significant
3	Freshwater Emergent	mean NDVI	Average Summer Temperature	0.065	0.7738	Weak Positive	Not Significant
3	Freshwater Emergent	mean NDVI	Average Winter-Spring Temperature	-0.1742	0.4382	Weak Negative	Not Significant
3	Freshwater Emergent	mean NDVI	Average Annual Temperature	-0.0642	0.7767	Weak Negative	Not Significant
3	Freshwater Emergent	mean NDVI	Cumulative Winter-Spring Precipitation	0.0217	0.9237	Weak Positive	Not Significant
3	Freshwater Emergent	mean NDVI	Annual Precipitation	0.1519	0.4997	Weak Positive	Not Significant
3	N/A	Winter-Spring Precipitation	Average Winter-Spring Temperature	-0.3472	0.1133	Moderate Negative	Not Significant
3	N/A	Annual Precipitation	Average Annual Temperature	-0.6231	0.0019	Strong Negative	Significant



Region	Wetland Type	variable 1	variable 2	Pearson Correlation	p-value	Correlation	Significance
4	Freshwater Forested	mean NDVI	Average Summer Temperature	0.0062	0.9781	Weak Positive	Not Significant
4	Freshwater Forested	mean NDVI	Average Winter-Spring Temperature	-0.2869	0.1955	Weak Negative	Not Significant
4	Freshwater Forested	mean NDVI	Average Annual Temperature	-0.1702	0.4489	Weak Negative	Not Significant
4	Freshwater Forested	mean NDVI	Cumulative Winter-Spring Precipitation	0.073	0.7469	Weak Positive	Not Significant
4	Freshwater Forested	mean NDVI	Annual Precipitation	0.3027	0.1709	Weak Positive	Not Significant
4	Freshwater Emergent	mean NDVI	Average Summer Temperature	0.0459	0.8394	Weak Positive	Not Significant
4	Freshwater Emergent	mean NDVI	Average Winter-Spring Temperature	-0.2172	0.3316	Weak Negative	Not Significant
4	Freshwater Emergent	mean NDVI	Average Annual Temperature	-0.0913	0.6861	Weak Negative	Not Significant
4	Freshwater Emergent	mean NDVI	Cumulative Winter-Spring Precipitation	0.062	0.7839	Weak Positive	Not Significant
4	Freshwater Emergent	mean NDVI	Annual Precipitation	0.2454	0.271	Weak Positive	Not Significant
4	N/A	Winter-Spring Precipitation	Average Winter-Spring Temperature	-0.1982	0.3766	Weak Negative	Not Significant
4	N/A	Annual Precipitation	Average Annual Temperature	-0.4263	0.0479	Moderate Negative	Significant



Figures 22 - 30 above contain tables showing the statistical results for each region. The tables show the climate region, wetland type, variable 1, variable 2, Pearson Correlation (r) value, p-value, correlation type (positive/negative), and whether the results were statistically significant (if p value was less than 0.05).

Region 1

In region 1, freshwater forested wetlands were analyzed for correlations between mean NDVI and various environmental variables. The correlation between mean NDVI and average summer temperature was weakly negative ($r = -0.0877$, $p = 0.698$), while the correlation with average Winter-Spring temperature was also weakly negative and non-significant ($r = -0.1805$, $p = 0.4216$). Similarly, the correlation with average annual temperature showed a weak negative relationship ($r = -0.1417$, $p = 0.5294$).

Conversely, mean NDVI and cumulative Winter-Spring precipitation showed a weak positive correlation ($r = 0.1966$, $p = 0.3805$). The correlation with annual precipitation was weakly positive but not statistically significant ($r = 0.2719$, $p = 0.229$).

For freshwater emergent wetlands in region 1, mean NDVI showed weak negative correlations with average summer temperature ($r = -0.0752$, $p = 0.7393$), average Winter-Spring temperature ($r = -0.1565$, $p = 0.4867$), and average annual temperature ($r = -0.1136$, $p = 0.6148$). The correlation with cumulative Winter-Spring precipitation was weakly positive but not statistically significant ($r = 0.1952$, $p = 0.3841$), like the correlation with annual precipitation ($r = 0.2709$, $p = 0.2227$).

In the N/A (climate variables) category of region 1, Winter-Spring precipitation and average Winter-Spring temperature exhibited a weakly negative and non-significant correlation ($r = -0.19$, $p = 0.397$). Conversely, the correlation between annual precipitation and average annual temperature was moderately negative and statistically significant ($r = -0.4429$, $p =$

0.039), indicating a notable relationship between these variables.

Region 2

In region 2, freshwater forested wetlands showed weak correlations between mean NDVI and environmental variables. The correlation with average summer temperature was weakly negative (r = -0.0647, p = 0.7747), and similarly, correlations with average Winter-Spring temperature and average annual temperature were weakly negative and non-significant (r = -0.2506, p = 0.2606 and r = -0.2031, p = 0.3646, respectively). The correlations with cumulative Winter-Spring precipitation and annual precipitation exhibited weak positive relationships, but they were not statistically significant (r = 0.0533, p = 0.8137 and r = 0.1385, p = 0.5388, respectively).

For freshwater emergent wetlands in region 2, the correlations between mean NDVI and environmental variables showed weak negative relationships without statistical significance. Specifically, mean NDVI and average summer temperature had an r value of -0.052, with a p-value of 0.8182. The correlations with average Winter-Spring temperature (-0.224, p = 0.3162) and average annual temperature (-0.187, p = 0.4048) were also weak and non-significant. In contrast, the correlation with cumulative Winter-Spring precipitation was weakly positive (r = 0.0398, p = 0.8604), and similarly, the correlation with annual precipitation was weakly positive (r = 0.1238, p = 0.5831), but neither were statistically significant.

In the N/A category of region 2, the correlations between Winter-Spring precipitation and average Winter-Spring temperature (r = -0.2082) and annual

precipitation and average annual temperature (r = -0.3024) were weakly negative and not statistically significant (p = 0.3525 and p = 0.1713, respectively).

Region 3

In region 3, freshwater forested wetlands exhibited weak correlations between mean NDVI and various environmental variables. Specifically, a weak positive correlation was observed between mean NDVI and Average Summer Temperature (r = 0.049, p = 0.8286), while weak negative correlations were found with Average Winter-Spring Temperature (r = -0.2926, p = 0.1864) and Average Annual Temperature (r = -0.1362, p = 0.5455). Similarly, weak positive correlations were identified with Cumulative Winter-Spring Precipitation (r = 0.0452, p = 0.8416) and Annual Precipitation (r = 0.234, p = 0.2947), but they were not statistically significant.

For freshwater emergent wetlands in region 3, mean NDVI showed weak positive correlations with Average Summer Temperature (r = 0.065, p = 0.7738) and Annual Precipitation (r = 0.1519, p = 0.4997), while weak negative correlations were observed with Average Winter-Spring Temperature (r = -0.1742, p = 0.4382) and Average Annual Temperature (r = -0.0642, p = 0.7767). Additionally, weak positive correlations with Cumulative Winter-Spring Precipitation (r = 0.0217, p = 0.9237) were not statistically significant.

In the N/A category of region 3, Winter-Spring Precipitation and Average Winter-Spring Temperature exhibited a moderately negative correlation (r = -0.3472, p = 0.1133), but it was not statistically significant. Conversely, Annual Precipitation and Average Annual Temperature showed a strong negative correlation (r = -0.6231, p = 0.0019), indicating statistical significance.

Region 4

In region 4, freshwater forested wetlands exhibited weak correlations between mean NDVI and environmental variables. The correlation with Average Summer Temperature was weakly positive (r = 0.0062, p = 0.9781), and similarly, correlations with Average Winter-Spring Temperature and Average Annual Temperature were weakly negative and non-significant (r = -0.2869, p = 0.1955 and r = -0.1702, p = 0.4489, respectively). The correlations with Cumulative Winter-Spring Precipitation and Annual Precipitation showed weak positive relationships, but they were not statistically significant (r = 0.073, p = 0.7469 and r = 0.3027, p = 0.1709, respectively).

For freshwater emergent wetlands in region 4, the correlations between mean NDVI and environmental variables displayed weak positive and negative relationships that were not statistically significant. Specifically, mean NDVI and Average Summer Temperature had an r value of 0.0459, with a p-value of 0.8394. The correlations with Average Winter-Spring Temperature (-0.2172, p = 0.3316) and Average Annual Temperature (-0.0913, p = 0.6861) were also weak and non-significant.

In contrast, the correlation with Cumulative Winter-Spring Precipitation was weakly positive (r = 0.062, p = 0.7839), and similarly, the correlation with Annual Precipitation was weakly positive (r = 0.2454, p = 0.271), but neither were statistically significant, except for Annual Precipitation and Average Annual Temperature, which showed a moderate negative correlation that was significant (r = -0.4263, p = 0.0479).

Discussion and Conclusion:

Previous research has shown that the relationship between temperature and NDVI, and precipitation and NDVI can vary depending on regional climate and latitude. For example, a 2010 study on global NDVI and climate trends showed that NDVI increases in northern and mid and high latitudinal zones are related to temperature rise, and NDVI decreases in southern semiarid regions are due to a precipitation decrease (Ichii, K; Kawabata, A; Yamaguchi, Y). Similar results were shown in a 1993 study (Schultz, P.A.; Halpert, M.S.). Data analyzed on a larger temporal scale has shown clear trends of increased precipitation, shifting distribution of annual precipitation toward Spring months, and increased temperatures in the Midwestern United States (Purdue University, 2018).

This study indicated weak positive relationships between mean NDVI for both forested and emergent wetlands when relating to cumulative Winter-Spring and annual precipitation across all regions, but the relationships were not shown to be significant. Additionally, region 3 and region 4 showed weak positive relationships between mean NDVI for both wetland types with average summer temperature. Again, the results for each of these were determined not to be significant.

In region 1, a significant and moderate negative relationship was observed between annual precipitation and average annual temperature. In region 3, another significant and strong negative relationship was observed for these variables. In region 4, another significant and moderate negative relationship was found between the same variables.

Given the established relationships between increased temperatures with increased moisture and precipitation, along

with the trends of steady increase for both precipitation and temperature shown by the Purdue University Climate Report from 2018, there is an interesting and slightly differing pattern being shown here on a more localized geographic and smaller temporal scale.

Of note is the different correlation results shown in regions 1 and 2 (Indiana), which indicated a weak negative relationship between mean NDVI and cumulative Winter-Spring precipitation, while regions 3 and 4 (Illinois) indicated a weak positive relationship between these variables. This brings into question whether differing drainage patterns, possibly resulting from channelization on the Indiana side, and no channelization on the Illinois side of the river, are having an influence on this relationship.

Possible future research could consider differences in floodplain boundaries, surface drainage and topography, and groundwater sources between Indiana and Illinois and how those relate to NDVI and climate variables. Other future research questions could analyze summer precipitation as it relates to NDVI, as there have been observed trends of summers becoming drier in Indiana.

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